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Workshop - 5

1. Create a function which returns a struct containing the area and perimeter of a circle. The function should take in the radius as it's only parameter.

Ans :

Source Code :

```
#include <stdio.h>

typedef struct{
 float area;
 float perimeter;
}attributes;

attributes calculate_area_and_perimeter(float r){
 attributes result;

 result.area=3.14*r*r;
 result.perimeter=2*3.14*r;

 return result;
}

int main(){
 float radius;
 attributes area_and_perimeter;

 printf("Enter the radius of the circle : ");
 scanf("%f",&radius);

 area_and_perimeter=calculate_area_and_perimeter(radius);

 printf("Radius = %f.\n",radius);
 printf("Circumference = %.2f. \n",area_and_perimeter.perimeter);
 printf("Area = %.2f. \n",area_and_perimeter.area);

 printf("\n");
 return 0;
}
```

Output :

```
Enter the radius of the circle : 101
Radius = 101.000000.
Circumference = 634.28.
Area = 32031.14.
```

2. Create a function which returns a struct containing two roots given from the quadratic formula. The function should take 3 integers as parameters named a, b and c.

Ans :

Source Code :

```
#include<stdio.h>
#include <math.h>

typedef struct{
 float root1;
 float root2;
}Roots;

Roots find_roots(int a,int b,int c){
 Roots r;
 float discriminant=b*b-4*a*c;

 if(discriminant>=0){
  r.root1=(-b+sqrt(discriminant))/(2*a);
  r.root2=(-b-sqrt(discriminant))/(2*a);
 }else{
  r.root1=r.root2=b*(2.0*a);
 }
 return r;
}

int main(){
 int a,b,c;
 Roots roots;

 printf("Enter values of a,b and c : ");
 scanf("%d %d %d",&a,&b,&c);

 roots=find_roots(a,b,c);

 printf("Root 1 = %.2f\n",roots.root1);
 printf("Root 2 = %.2f\n",roots.root2);

 return 0;
}
```

Output :

```
Enter values of a,b and c : 1 5 6
Root 1 = -2.00
Root 2 = -3.00
```

3. Recreate question 2 but returning a pointer (float). Use the example given in the lecture on how to return a pointer to an array of a specific variable type.

Ans :

Source Code :

```
#include<stdio.h>
#include <math.h>
#include <stdlib.h>

float* find_roots(int a,int b,int c){
 float *roots=(float*) malloc(2*sizeof(float));
 float discriminant=b*b-4*a*c;

 if(discriminant>=0){
  roots[0]=(-b+sqrt(discriminant))/(2*a);
  roots[1]=(-b-sqrt(discriminant))/(2*a);
 }else{
  roots[0]=roots[0]=b*(2.0*a);
 }
 return roots;
}

int main(){
 int a,b,c;
 float *roots;

 printf("Enter values of a,b and c : ");
 scanf("%d %d %d",&a,&b,&c);

 roots=find_roots(a,b,c);

 printf("Root 1 = %.2f\n",roots[0]);
 printf("Root 2 = %.2f\n",roots[1]);

 free(roots);

 return 0;
}
```

Output :

```
Enter values of a,b and c : 1 5 6
Root 1 = -2.00
Root 2 = -3.00
```

4. Write a structure to store the names, salary, and hours of work per day of 5 employees in a company. Write a program to increase the salary depending on the number of hours of work per day as follows and then print the name of

all the employees along with their final salaries. To do this, you should create employees within the main, then check whether their hours fit within the boundaries below, if so, increase the salary.

Hours of work per day	8	10	>=12
Increase in salary	$50	$100	$150

Ans :

Source Code :

```
#include <stdio.h>

typedef struct{
 char name[20];
 int salary;
 int work_hours;
} Employee;

int increase_salary(int hours,int salary){
 if(hours>=12){
  return salary+150;
 }else if(hours>=10){
  return salary+100;
 }else if(hours>=8){
  return salary+50;
 }else{
  return salary;
 }
}

int main(){
 Employee e[5];
 int new_salary;

 for(int i=0;i<5;i++){
  printf("Enter name,salary,work hours for Employee No. %d : ",i+1);
  scanf(" %[^,],%d,%d",e[i].name,&e[i].salary,&e[i].work_hours);
 }

 printf("\nEmployee Details : \n");

 for(int i=0;i<5;i++){
  new_salary=increase_salary(e[i].work_hours,e[i].salary);
  printf("Name : %s\nOld Salary : %d\nWork Hours : %d\nNew Salary : %d\n",e[i].name,e[i].salary,e[i].work_hours,new_salary);
 }


 printf("\n");
 return 0;
}
```

Output :

```
Enter name,salary,work hours for Employee No. 1 : John,12000,14
Enter name,salary,work hours for Employee No. 2 : Jeffrey,20000,11
Enter name,salary,work hours for Employee No. 3 : Jason,7000,9
Enter name,salary,work hours for Employee No. 4 : Jamie,27000,6
Enter name,salary,work hours for Employee No. 5 : Joffrey,10000,12

Employee Details :
Name : John
Old Salary : 12000
Work Hours : 14
New Salary : 12150
Name : Jeffrey
Old Salary : 20000
Work Hours : 11
New Salary : 20100
Name : Jason
Old Salary : 7000
Work Hours : 9
New Salary : 7050
Name : Jamie
Old Salary : 27000
Work Hours : 6
New Salary : 27000
Name : Joffrey
Old Salary : 10000
Work Hours : 12
New Salary : 10150
```

5. Complete question 4 but using a separate function and pass in the struct which contains the employee information, this will be a pointer. You only need to pass in one struct, not multiple structs (i.e., only one employee). You will need to think back to the lecture regarding pointers to change the original values from the original address where the employee was created.

Ans :

Source Code :

```c
#include <stdio.h>

typedef struct{
 char name[20];
 int salary;
 int work_hours;
} Employee;

int increase_salary(Employee *e){
 if(e->work_hours>=12){
  return e->salary+=150;
 }else if(e->work_hours>=10){
  return e->salary+=100;
 }else if(e->work_hours>=8){
  return e->salary+=50;
 }
}

int main(){
 Employee e[5];
 int old_salary;

 for(int i=0;i<5;i++){
        printf("Enter name,salary,work_hours for Employee No. %d : ",i+1);
        scanf(" %[^,],%d,%d",e[i].name,&e[i].salary,&e[i].work_hours);
 }

 printf("\nEmployee Details : \n");

 for(int i=0;i<5;i++){
  old_salary=e[i].salary;
  increase_salary(&e[i]);
  printf("Name : %s\nOld Salary : %d\nWork Hours : %d\nNew Salary : %d\n",e[i].name,old_salary,e[i].work_hours,e[i].salary);
 }


 printf("\n");
 return 0;
}
```

Output :

```
Enter name,salary,work_hours for Employee No. 1 : John,12000,14
Enter name,salary,work_hours for Employee No. 2 : Jeffrey,20000,11
Enter name,salary,work_hours for Employee No. 3 : Jason,7000,9
Enter name,salary,work_hours for Employee No. 4 : Jamie,27000,6
Enter name,salary,work_hours for Employee No. 5 : Joffrey,10000,12

Employee Details :
Name : John
Old Salary : 12000
Work Hours : 14
New Salary : 12150
Name : Jeffrey
Old Salary : 20000
Work Hours : 11
New Salary : 20100
Name : Jason
Old Salary : 7000
Work Hours : 9
New Salary : 7050
Name : Jamie
Old Salary : 27000
Work Hours : 6
New Salary : 27000
Name : Joffrey
Old Salary : 10000
Work Hours : 12
New Salary : 10150
```

6. Let us work on the menu of a library. Create a structure containing book information like accession number, name of author, book title and flag to know whether book is issued or not.
   Create a menu in which the following can be done.
   1 - Display book information
   2 - Add a new book
   3 - Display all the books in the library of a particular author
   4 - Display the number of books of a particular title
   5 - Display the total number of books in the library
   6 - Issue a book
   (If we issue a book, then its number gets decreased by 1 and if we add a book, its number gets increased by 1).

Ans :

Source Code :

```c
#include <stdio.h>
#include <string.h>

typedef struct{
        int accession_number;
        char author[50];
        char title[50];
        int number_of_copies;
} Book;

int main(){
        Book b[5];
        int n=0;
        int choice;

        while(1){
                printf("\nMake your choice.\n");
                printf("1. Display book information\n");
    printf("2. Add a new book\n");
    printf("3. Display all the books in the library of a particular author\n");
    printf("4. Display the number of books of a particular title\n");
    printf("5. Display the total number of books in the library\n");
    printf("6. Issue a book\n");
    printf("7. Exit\n");
    printf("Enter choice: ");
    scanf("%d",&choice);

                if(choice==1){
        for(int i=0;i<n;i++){
                    printf("\nAccession No: %d\n", b[i].accession_number);
                    printf("Author: %s\n",b[i].author);
                    printf("Title: %s\n",b[i].title);
                    printf("Copies: %d\n",b[i].number_of_copies);
            }
      }
    else if(choice==2){
        printf("Enter accession no: ");
        scanf("%d",&b[n].accession_number);

        printf("Enter author: ");
        scanf(" %[^\n]",b[n].author);

        printf("Enter title: ");
        scanf(" %[^\n]",b[n].title);

        printf("Enter number of copies: ");
        scanf("%d", &b[n].number_of_copies);
                                n++;
        }
    else if(choice==3){
        char search_author[50];
        printf("Enter author name: ");
        scanf(" %[^\n]",search_author);
```

```
                for(int i=0;i<n;i++){
            if(strcmp(b[i].author,search_author)==0){
                printf("\nTitle: %s (Copies: %d)\n",b[i].title, b[i].number_of_copies);
                }
            }
        }
    else if(choice==4){
         char search_title[50];
         int total=0;

         printf("Enter title: ");
         scanf(" %[^\n]",search_title);
         for(int i = 0; i < n; i++){
          if(strcmp(b[i].title,search_title)==0){
             total+=b[i].number_of_copies;
                }
            }
          printf("Total copies of '%s': %d\n",search_title,total);
        }
    else if(choice==5){
         int total = 0;
         for(int i=0;i<n;i++){
            total+=b[i].number_of_copies;
           }
            printf("Total books in library: %d\n",total);
        }
    else if(choice==6){
        int acc;
        printf("Enter accession number to issue: ");
        scanf("%d",&acc);
        for(int i=0;i<n;i++){
           if(b[i].accession_number==acc){
             if(b[i].number_of_copies>0){
               b[i].number_of_copies--;
               printf("Book issued successfully.\n");
              }else{
                printf("No copies available.\n");
                    }
                }
            }
        }
    else if(choice==7){
        break;
   }
    else{
        printf("Invalid choice.\n");
      }
    }
        printf("\n");
        return 0;
}
```

Output :

```
Make your choice.
1. Display book information
2. Add a new book
3. Display all the books in the library of a particular author
4. Display the number of books of a particular title
5. Display the total number of books in the library
6. Issue a book
7. Exit
Enter choice: 2
Enter accession no: 1
Enter author: Jack Marston
Enter title: Red Dead Redemption
Enter number of copies: 6

Make your choice.
1. Display book information
2. Add a new book
3. Display all the books in the library of a particular author
4. Display the number of books of a particular title
5. Display the total number of books in the library
6. Issue a book
7. Exit
Enter choice: 2
Enter accession no: 2
Enter author: J.K. Rowling
Enter title: Harry Potter and the Sorcerer's Stone
Enter number of copies: 5

Make your choice.
1. Display book information
2. Add a new book
3. Display all the books in the library of a particular author
4. Display the number of books of a particular title
5. Display the total number of books in the library
6. Issue a book
7. Exit
Enter choice: 2
Enter accession no: 3
Enter author: J.K. Rowling
Enter title: Harry Potter and the Chamber of Secrets
Enter number of copies: 3

Make your choice.
1. Display book information
2. Add a new book
3. Display all the books in the library of a particular author
4. Display the number of books of a particular title
5. Display the total number of books in the library
6. Issue a book
7. Exit
Enter choice: 2
Enter accession no: 4
```

```
Enter author: George Orwell
Enter title: 1984
Enter number of copies: 4

Make your choice.
1. Display book information
2. Add a new book
3. Display all the books in the library of a particular author
4. Display the number of books of a particular title
5. Display the total number of books in the library
6. Issue a book
7. Exit
Enter choice: 5
Total books in library: 18

Make your choice.
1. Display book information
2. Add a new book
3. Display all the books in the library of a particular author
4. Display the number of books of a particular title
5. Display the total number of books in the library
6. Issue a book
7. Exit
Enter choice: 1

Accession No: 1
Author: Jack Marston
Title: Red Dead Redemption
Copies: 6

Accession No: 2
Author: J.K. Rowling
Title: Harry Potter and the Sorcerer's Stone
Copies: 5

Accession No: 3
Author: J.K. Rowling
Title: Harry Potter and the Chamber of Secrets
Copies: 3

Accession No: 4
Author: George Orwell
Title: 1984
Copies: 4

Make your choice.
1. Display book information
2. Add a new book
3. Display all the books in the library of a particular author
4. Display the number of books of a particular title
5. Display the total number of books in the library
6. Issue a book
7. Exit
```

```
Enter choice: 3
Enter author name: J.K. Rowling

Title: Harry Potter and the Sorcerer's Stone (Copies: 5)

Title: Harry Potter and the Chamber of Secrets (Copies: 3)

Make your choice.
1. Display book information
2. Add a new book
3. Display all the books in the library of a particular author
4. Display the number of books of a particular title
5. Display the total number of books in the library
6. Issue a book
7. Exit
Enter choice: 4
Enter title: Harry Potter and the Sorcerer's Stone
Total copies of 'Harry Potter and the Sorcerer's Stone': 5

Make your choice.
1. Display book information
2. Add a new book
3. Display all the books in the library of a particular author
4. Display the number of books of a particular title
5. Display the total number of books in the library
6. Issue a book
7. Exit
Enter choice: 5
Total books in library: 18

Make your choice.
1. Display book information
2. Add a new book
3. Display all the books in the library of a particular author
4. Display the number of books of a particular title
5. Display the total number of books in the library
6. Issue a book
7. Exit
Enter choice: 6
Enter accession number to issue: 3
Book issued successfully.

Make your choice.
1. Display book information
2. Add a new book
3. Display all the books in the library of a particular author
4. Display the number of books of a particular title
5. Display the total number of books in the library
6. Issue a book
7. Exit
Enter choice: 8
Invalid choice.

Make your choice.
1. Display book information
2. Add a new book
3. Display all the books in the library of a particular author
4. Display the number of books of a particular title
5. Display the total number of books in the library
6. Issue a book
7. Exit
Enter choice: 7
```

7. In the lecture you were shown how to use arrays and malloc (). Write a function fibonacci(int size) that returns a dynamically allocated array containing the first “size” Fibonacci numbers. For example, if you call fibonacci(5), it should return an array containing: 0, 1, 1, 2, 3.

Ans :

Source Code :

```c
#include <stdio.h>
#include <stdlib.h>

int* fibonacci(int size){
    int *arr=(int*)malloc(size*sizeof(int));

    if(size>=1)arr[0]=0;
    if(size>=2)arr[1]=1;

    for(int i=2;i<size;i++){
        arr[i]=arr[i-1]+arr[i-2];
    }
    return arr;
}

int main(){
    int n;
    printf("Enter how many elements: ");
    scanf("%d",&n);

    int *fib=fibonacci(n);

    if(fib!=NULL){
        printf("Fibonacci series upto %d terms : \n",n);
        for(int i=0;i<n;i++){
            printf("%d ",fib[i]);
        }
        free(fib);
    }

    printf("\n");
    return 0;
}
```

Output :

```
Enter how many elements: 10
Fibonacci series upto 10 terms :
0 1 1 2 3 5 8 13 21 34
```

8. Write a function called primes(int n) that returns a dynamically allocated array containing the first n prime numbers. For example, if you call primes(5), it should return an array containing: 2, 3, 5, 7, 11.

Ans :

Source Code :

```c
#include <stdio.h>
#include <stdlib.h>

int prime_check(int num){
    if(num<=1) return 0;

    for(int i=2;i*i<=num;i++){
        if(num%i==0)
                return 0;
    }
    return 1;
}

int* primes(int n){
    int *arr=(int*)malloc(n*sizeof(int));
    int count=0;
    int num=2;

    while(count<n){
        if(prime_check(num)){
            arr[count]=num;
            count++;
        }
        num++;
    }
    return arr;
}

int main(){
    int n;
    printf("Enter how many prime numbers you want : ");
    scanf("%d",&n);

    int *p=primes(n);
    if(p!=NULL){
        printf("First %d prime numbers:\n",n);
        for(int i=0;i<n;i++){
            printf("%d ",p[i]);
        }
        free(p);
    }

    printf("\n");
    return 0;
}
```

Output :

```
Enter how many prime numbers you want : 8
First 8 prime numbers:
2 3 5 7 11 13 17 19
```

9. Write a program that asks the user for n (how many numbers they want to store). Use malloc to create an integer array and let the user enter n numbers into it. Then use malloc again to create a second integer array of the same size, and copy the numbers into it in reverse order. Print and then free() both arrays.

Ans :
Source Code :

```
#include <stdio.h>
#include <stdlib.h>

int main(){
    int n;

    printf("Enter how many numbers you want to store : ");
    scanf("%d",&n);

    int *integer_arr_1=(int*) malloc(n*sizeof(int));

    printf("Enter %d numbers:\n",n);
    for(int i=0;i<n;i++){
        scanf("%d",&integer_arr_1[i]);
    }

    int *integer_arr_2=(int*)malloc(n*sizeof(int));

    for(int i=0;i<n;i++){
        integer_arr_2[i]=integer_arr_1[n-1-i];
    }

    printf("\nArray 1 : \n");
    for(int i=0;i<n;i++){
        printf("%d ",integer_arr_1[i]);
    }

    printf("\nArray 1 reversed : \n");
    for(int i=0;i<n;i++){
        printf("%d ",integer_arr_2[i]);
    }

    free(integer_arr_1);
    free(integer_arr_2);

    printf("\n");
    return 0;
}
```

Output :

```
Enter how many numbers you want to store : 5
Enter 5 numbers:
1
2
3
4
5

Array 1 :
1 2 3 4 5
Array 1 reversed :
5 4 3 2 1
```